

Assessment Cover Sheet

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ASSESSMENT DETAILS

Unit title	COMPUTING TECHNOLOGY INNOVATION PROJECT	Tutorial /Lab Group	-	Office use only
Unit code	COS 30049	Due date	17/4/2026	
Name of lecturer/tutor	Ts. Dr. Lee Su han			
Assignment title	Sprint 1 Report			Faculty or school date stamp

STUDENT(S) DETAILS

	Student Name(s)	Student ID Number(s)
(1)	TERENCE CHIN SENG WONG	104404059
(2)	TAMMY RU XIU TAY	104405861
(3)	MAIAW FONG LIM	104406013
(4)	EMILY HUI XIU JONG	102788832
(5)	ZI YANG LIM	102774129
(6)	YOUNG FUNG LAU	102774585
(7)	BRIAN JIA HUA LIEW	104381288

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(3)	LIM	(6)	LIEW
(7)	JONG		

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Sprint 1 Report

Digital Park Guide Training Platform Sarawak Forestry Corporation (SFC)

Unit Code	COS30049
Sprint	Sprint 1
Sprint Duration	9 March 2026 — 17 April 2026 (6 weeks)
Working Cadence	3 days/week, rotating pairs, 6 hrs/session
Total Person-Hours	~216 hrs (6 weeks × 3 days × 2 members × 6 hrs)
Supervisor / Client	Ts. Dr. Lee Su Han

Project Team

#	Name	Student ID	Role
1	Terence Chin Seng Wong	104404059	Project Lead, Full Stack Lead, AI & Media Engineer
2	Tammy Ru Xiu Tay	104405861	Full Stack Developer, AI & Media Engineer
3	Miaw Fong Lim	104406013	Frontend Developer
4	Emily Hui Xiu Jong	102788832	Frontend Developer
5	Zi Yang Lim	102774129	Full Stack Developer, AI & Media Engineer
6	Young Fung Lau	102774585	Backend Developer, QA & Security Engineer
7	Brian Jia Hua Liew	104381288	Backend Developer, QA & Security Engineer

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1. Sprint 1 Backlog Items

Sprint 1 covers Tracks A (Authentication & Access Control), B1–B3 (core training and quiz), C (planned but deferred to Sprint 2), and the AI monitoring prototype (D1–D4). Tracks B4–B8, C, D5–D7, E, and F remain as Sprint 2 items.

The sprint backlog below reflects the planned items, their subtasks, business value, and actual completion status as of 17 April 2026.

No.	Item	Subtask	Functionality	BV	Status
A1	User Auth (OAuth + OTP)	A1.1	Configure Google OAuth in Supabase Auth	10	Done
		A1.2	Implement Google sign-in button and OAuth redirect handler		Done
		A1.3	Implement OTP request form, trigger, and verification UI		Done
		A1.4	Configure session listener, token refresh, and cookie storage		Done
		A1.5	Implement error states for invalid OTP, expired session, OAuth failure		Done
A2	Role-Based Access Control	A2.1	Define role enum (SUPERADMIN, HOD, GUIDE, PUBLIC) and permission matrix	10	Done
		A2.2	Write and apply RLS policies on all tables scoped per role		Done
		A2.3	Implement Next.js middleware for JWT role claim and redirect		Done
		A2.4	Implement client-side role guards for conditional UI rendering		Done
		A2.5	Write cross-role test cases verifying data boundary enforcement		Done
A3	Public Browsing	A3.1	Build unauthenticated announcements listing page	7	In Progress
		A3.2	Build public training programme brief listing per TPA		In Progress
		A3.3	Configure public Next.js routes, metadata tags, Open Graph data		In Progress
B1	Training Module Upload (HoD)	B1.1	Implement CRUD API routes for training modules with HoD role guard	9	Done
		B1.2	Implement file upload to Supabase Storage with size validation		Done
		B1.3	Build HoD module editor with rich text, file attachment, TPA selector		Done
		B1.4	Tag each module to a specific TPA and certification track		Done

No.	Item	Subtask	Functionality	BV	Status
		B1.5	Track module edit history with timestamps for audit and rollback		Done
B2	Training Module View (Guide)	B2.1	Build guide module listing with progress indicator and locked state	9	Done
		B2.2	Implement video player and PDF viewer for module content		Done
		B2.3	Record and update guide progress per module on consumption		In Progress
		B2.4	Trigger completion state on all content consumed; update badge eligibility		In Progress
B3	Quiz System	B3.1	Build HoD quiz builder with MCQ editor and per-module assignment	8	Done
		B3.2	Build guide quiz interface with navigation, timer, and submit		Done
		B3.3	Calculate and persist quiz score; flag pass/fail per threshold		In Progress
		B3.4	Show result summary; allow retry with configurable attempt limit		In Progress
B4	Certification Application	B4.1–B4.4	Application form, submission API, status tracking, document upload	9	Not Started
B5	HoD Review & Interview	B5.1–B5.4	Review dashboard, approve/reject, interview scheduling, email notification	9	Not Started
B6	Stripe Payment (Sandbox)	B6.1–B6.5	Stripe SDK setup, payment session, webhook handler, retry flow, status display	7	Not Started
B7	Badge / Certificate Issuance	B7.1–B7.4	Badge DB insert, PDF generation, email delivery, badge display	9	Not Started
B8	Senior Auto-Nomination	B8.1–B8.4	Edge Function badge check, nomination criteria, HoD notification, role promotion	6	Not Started
C1	Email Notifications	C1.1–C1.3	Supabase email triggers, HTML templates, delivery logging	8	Not Started
C2	SMS Notifications (Twilio)	C2.1–C2.3	Twilio SDK setup, SMS API route, delivery status logging	7	Not Started
C3	In-App Notifications (Realtime)	C3.1–C3.3	Supabase Realtime subscription, notification bell UI, read/unread state	8	Not Started
D1	PWA Service Worker	D1.1–D1.4	Service worker registration, model caching, cache invalidation, offline assets	9	In Progress
D2	Camera Feed	D2.1–D2.4	getUserMedia, stream to canvas, session controls, frame extraction	9	Done
D3	On-Device Inference	D3.1–D3.4	ONNX Runtime Web, model load, frame inference loop, confidence threshold	10	Done
D4	On-Device Alert	D4.1–D4.3	Web Audio API beep, visual overlay, dismiss and session continuation	8	Done

2. Work Breakdown Structure (WBS) — Sprint 1

The WBS below breaks down all Sprint 1 work items into subtasks with estimated hours and primary assignees. The sprint total is estimated at 145 active development hours across all tasks. The AI monitoring track (D2–D4 and model training) was completed ahead of its Sprint 2 schedule and is included here as a Sprint 1 achievement.

Column legend: ✓ = primary contributor for that subtask.

ID	Task / Subtask	Hours	Terence	Tammy	Zi Yang	Brian	Emily	Miaw Fong	Young Fung
A1	User Registration & Authentication	16 hrs							
A1.1	Google OAuth provider setup	3 hrs	✓						
A1.2	OAuth sign-in flow (Next.js)	4 hrs	✓						
A1.3	OTP email/phone verification flow	4 hrs	✓						
A1.4	JWT session management	3 hrs	✓						
A1.5	Auth error handling & user feedback	2 hrs	✓						
A2	Role-Based Access Control (RBAC)	20 hrs							
A2.1	Role enum & permission matrix	3 hrs	✓	✓					
A2.2	PostgreSQL RLS policies	6 hrs	✓	✓					
A2.3	Next.js middleware route protection	3 hrs	✓	✓			✓	✓	
A2.4	Role-based UI conditional rendering	4 hrs	✓						
A2.5	RLS policy unit testing	4 hrs			✓	✓	✓	✓	✓
A3	Public Browsing (Placeholder)	4 hrs							
A3.1–3	Announcements, programme briefs, route config	4 hrs		✓			✓	✓	✓
B1	Training Module Upload & Management (HoD)	19 hrs							

ID	Task / Subtask	Hours	Terence	Tammy	Zi Yang	Brian	Emily	Miaw Fong	Young Fung
B1.1	Module CRUD API routes	4 hrs			✓				
B1.2	Media file upload to Supabase Storage	3 hrs			✓				
B1.3	Module editor UI (HoD dashboard)	6 hrs			✓				
B1.4	TPA-specific module categorisation	3 hrs			✓				
B1.5	Module versioning & update history	3 hrs			✓				
B2	Training Module View & Completion (Guide)	13 hrs							
B2.1	Module listing with progress indicator	4 hrs			✓				✓
B2.2	Content player component	5 hrs			✓				✓
B2.3	Progress state persistence (in progress)	2 hrs			✓				✓
B2.4	Module completion marking (in progress)	2 hrs			✓				✓
B3	Quiz System	13 hrs							
B3.1	Quiz builder (HoD)	5 hrs				✓			
B3.2	Quiz attempt UI (Guide)	4 hrs				✓			
B3.3	Result scoring & storage (in progress)	2 hrs				✓			
B3.4	Retry and result display logic (in progress)	2 hrs				✓			
D1	PWA Service Worker & Model Caching (in progress)	6 hrs							
D1.1–4	Service worker lifecycle, model cache-first, invalidation, offline assets	6 hrs	✓	✓					
D2	Device Camera Initialisation & Live Feed	8 hrs							

ID	Task / Subtask	Hours	Terence	Tammy	Zi Yang	Brian	Emily	Miaw Fong	Young Fung
D2.1–4	getUserMedia, stream to canvas, session controls, frame extraction	8 hrs	✓	✓					
D3	On-Device Inference (ONNX Runtime Web)	18 hrs							
D3.1–2	ONNX Runtime Web integration, model load	6 hrs	✓	✓					
D3.3	Frame-by-frame inference loop with bounding box overlay	8 hrs	✓	✓					
D3.4	Confidence threshold, NMS, class filtering	4 hrs	✓	✓					
D4	On-Device Alert (Sound & Visual)	6 hrs							
D4.1	Web Audio API beep on detection trigger	2 hrs	✓	✓					
D4.2	Visual alert overlay component	2 hrs	✓	✓					
D4.3	Alert dismiss & session continuation	2 hrs	✓						
AI	YOLOv8n Model Training & Dataset Pipeline	22 hrs							
AI.1	Dataset collection from Roboflow (hand, person, wildlife, flower)	8 hrs	✓	✓	✓				
AI.2	Dataset merge, class remapping, augmentation in Roboflow	4 hrs	✓	✓	✓				
AI.3	YOLOv8n fine-tuning (100 epochs, early stop at 72)	6 hrs	✓	✓					
AI.4	ONNX export, browser integration testing, threshold tuning	4 hrs	✓						

3. Task Tracking — Trello Board

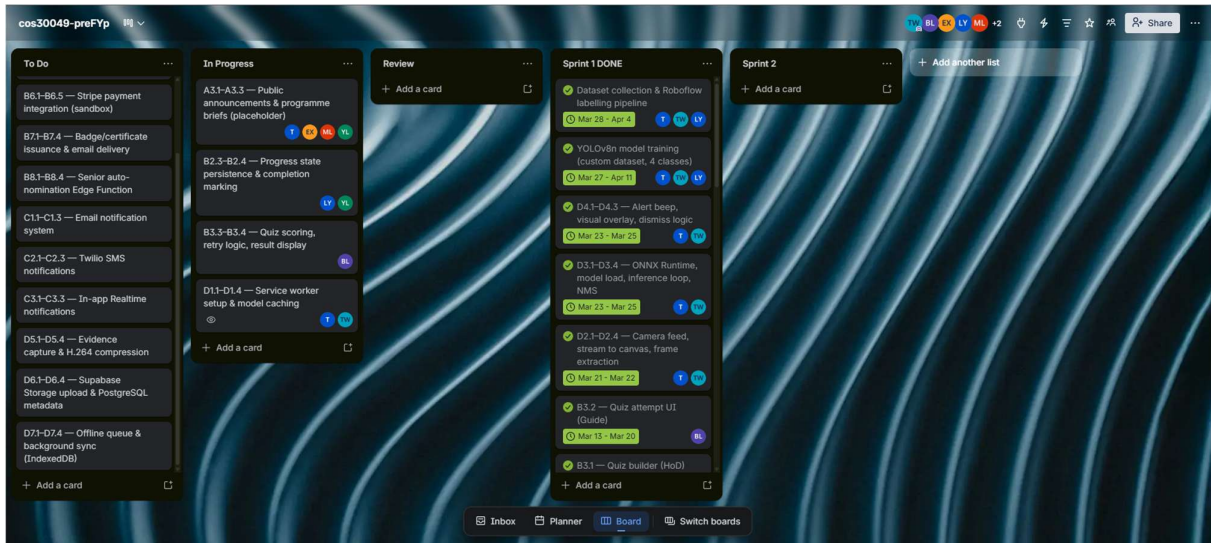
The following table represents the complete task list to be entered into the Trello board for Sprint 1. Tasks are organised across three lists: TO DO, DOING, and DONE. Each card includes the task name, assigned team member(s), estimated hours, and current status.

Note: The team will set up burndownfortrello.com using the hours listed below. Please insert the Trello board screenshot and burndown chart below the task table once the board has been configured.

Task Name	Assigned To	Hrs	Status
A1.1 — Google OAuth provider setup	Terence	3	Done
A1.2 — OAuth sign-in flow (Next.js)	Terence	4	Done
A1.3 — OTP email/phone verification flow	Terence	4	Done
A1.4 — JWT session management	Terence	3	Done
A1.5 — Auth error handling & user feedback	Terence	2	Done
A2.1 — Role enum & permission matrix	Terence, Tammy	3	Done
A2.2 — PostgreSQL RLS policies	Terence, Tammy	6	Done
A2.3 — Next.js middleware route protection	Terence, Tammy, Emily, Miaw Fong	3	Done
A2.4 — Role-based UI conditional rendering	Terence	4	Done
A2.5 — RLS policy unit testing	Zi Yang, Brian, Emily, Miaw Fong, Young Fung	4	Done
B1.1 — Module CRUD API routes	Zi Yang	4	Done
B1.2 — Media file upload to Supabase Storage	Zi Yang	3	Done
B1.3 — Module editor UI (HoD dashboard)	Zi Yang	6	Done
B1.4 — TPA-specific module categorisation	Zi Yang	3	Done
B1.5 — Module versioning & update history	Zi Yang	3	Done
B2.1 — Module listing with progress indicator	Zi Yang, Young Fung	4	Done
B2.2 — Content player component (video + PDF)	Zi Yang, Young Fung	5	Done
B3.1 — Quiz builder (HoD)	Brian	5	Done

Task Name	Assigned To	Hrs	Status
B3.2 — Quiz attempt UI (Guide)	Brian	4	Done
D2.1–D2.4 — Camera feed, stream to canvas, frame extraction	Terence, Tammy	8	Done
D3.1–D3.4 — ONNX Runtime, model load, inference loop, NMS	Terence, Tammy	18	Done
D4.1–D4.3 — Alert beep, visual overlay, dismiss logic	Terence, Tammy	6	Done
YOLOv8n model training (custom dataset, 4 classes)	Terence, Tammy, Zi Yang	12	Done
Dataset collection & Roboflow labelling pipeline	Terence, Tammy, Zi Yang	10	Done
A3.1–A3.3 — Public announcements & programme briefs (placeholder)	Tammy, Emily, Miaw Fong, Young Fung	4	In Progress
B2.3–B2.4 — Progress state persistence & completion marking	Zi Yang, Young Fung	4	In Progress
B3.3–B3.4 — Quiz scoring, retry logic, result display	Brian	4	In Progress
D1.1–D1.4 — Service worker setup & model caching	Terence, Tammy	6	In Progress
B4.1–B4.4 — Certification application submission	TBD	12	Not Started
B5.1–B5.4 — HoD review, interview scheduling, email notification	TBD	12	Not Started
B6.1–B6.5 — Stripe payment integration (sandbox)	TBD	10	Not Started
B7.1–B7.4 — Badge/certificate issuance & email delivery	TBD	10	Not Started
B8.1–B8.4 — Senior auto-nomination Edge Function	TBD	8	Not Started
C1.1–C1.3 — Email notification system	TBD	8	Not Started
C2.1–C2.3 — Twilio SMS notifications	TBD	6	Not Started
C3.1–C3.3 — In-app Realtime notifications	TBD	8	Not Started
D5.1–D5.4 — Evidence capture & H.264 compression	TBD	10	Not Started
D6.1–D6.4 — Supabase Storage upload & PostgreSQL metadata	TBD	8	Not Started
D7.1–D7.4 — Offline queue & background sync (IndexedDB)	TBD	10	Not Started

3.1 Trello Board Screenshot



3.2 Burndown Chart

The burndown chart is generated via burndownfortrello.com using the sprint hours defined in the Trello board above. The ideal burndown assumes a linear reduction from 145 hours on Day 1 to 0 hours by the end of Week 6 (17 April 2026).



Note on Burndown Chart Shape

The burndown chart reflects the actual completion timestamps recorded by Trello when cards were moved to the "Sprint 1 DONE" list. As most cards were verified and formally marked as done in a batch near the end of the sprint review period (April 13–17), the "Hours Completed" line appears to rise steeply at that point rather than incrementally.

In practice, development work was carried out progressively throughout the sprint — camera feed and inference pipeline (D2–D4) were completed March 21–25, model training ran March 27 – April 4, and UI components (B3.1–B3.2) were completed March 13–20. The steep curve is an artifact of batch card management rather than late delivery, and the team's actual progress aligns with the sprint plan. At the start of Sprint 2, or 20 April onwards, the team will acknowledge this problem and make sure to update Trello card daily to make sure this will no happen again in neither Sprint 2 report nor Final report submission Trello burn down chart.

4. Sprint 1 Progress Summary**4.1 Completed Items**

The following backlog items were fully completed during Sprint 1:

- A1 — Full authentication system implemented by Terence with Google OAuth and Supabase OTP. JWT session management, token refresh, and error handling for invalid OTP and OAuth failure are all working.
- A2 — Role-Based Access Control implemented for Superadmin, HoD, Guide, and Public roles. PostgreSQL RLS policies authored by Terence and Tammy; middleware route protection by Terence, Tammy, Emily, and Miaw Fong; cross-role unit testing by Zi Yang, Brian, Emily, Miaw Fong, and Young Fung.
- B1 — Training module management fully functional for HoD. All five subtasks completed by Zi Yang: CRUD API routes, Supabase Storage upload, module editor UI, TPA categorisation, and module versioning.
- B2 (UI) — Guide-facing module listing and content player implemented by Zi Yang with backend support from Young Fung. Progress state persistence and completion marking are in progress.
- B3 (partial) — Quiz builder and quiz attempt UI completed by Brian. Scoring logic and retry flow are in progress.

- D2 — Live webcam feed successfully piped to HTML canvas by Terence and Tammy, with session controls and configurable frame extraction at 15 FPS.
- D3 — On-device inference fully working using ONNX Runtime Web (WebAssembly) implemented by Terence and Tammy. Custom YOLOv8n model trained on 26,496 images across 4 classes achieving mAP50 of 0.807 overall. Real-time bounding box overlay renders directly on the video feed.
- D4 — On-device alert system implemented by Terence and Tammy. Web Audio API beep, visual overlay, and alert dismissal are all functional.
- AI Model Training — Dataset collection and Roboflow pipeline by Terence, Tammy, and Zi Yang. YOLOv8n fine-tuned by Terence and Tammy on RTX 5070 Ti, 72 epochs (early stop at patience 15). Per-class results: hand 0.992, wildlife 0.914, person 0.720, flower 0.603 mAP50. ONNX export and threshold tuning by Terence.

4.2 In-Progress Items

The following items were started but not fully completed by end of Sprint 1:

- A3 — Public browsing pages (Tammy, Emily, Miaw Fong, Young Fung) exist as structural placeholders. Not yet populated with real content — low priority as it depends on HoD-entered data.
- B2.3–B2.4 — Progress state persistence and module completion marking (Zi Yang, Young Fung) are implemented in the UI but backend PostgreSQL persistence is still being finalised.
- B3.3–B3.4 — Quiz scoring and retry logic (Brian) are partially implemented. Pass/fail threshold configuration and retry limit enforcement are still in progress.
- D1 — Service worker setup (Terence, Tammy) is in progress. The model currently loads from /public/models/yolov8n.onnx without service worker caching. Cache-first strategy and offline asset configuration will be completed in Sprint 2.

4.3 Deferred to Sprint 2

The following Sprint 1 planned items were deferred to Sprint 2 without negative impact on overall project progress:

- B4–B8 — Certification application, HoD review, Stripe payment, badge issuance, and senior auto-nomination. These depend on B1–B3 being fully stable, which is now the case.
- C1–C3 — Email, SMS, and in-app notification systems. Infrastructure is architected in the database schema but delivery mechanisms have not been wired up yet.
- D5–D7 — Evidence capture, H.264 compression via FFmpeg.wasm, Supabase Storage upload, and offline queue. The detection trigger is working; the evidence pipeline is the primary Sprint 2 AI deliverable.

4.4 Key Technical Achievement — AI Monitoring

The AI monitoring prototype was completed significantly ahead of its planned Sprint 2 timeline. Key achievements:

- Custom YOLOv8n model trained on domain-specific dataset (EgoHands, wildlife poaching, hand detection, outdoor flower datasets) assembled and labelled by Terence, Tammy, and Zi Yang via Roboflow.
- Two-stage violation detection: Stage 1 detects hand, person, wildlife, flower; Stage 2 checks IoU overlap between hand box and violation targets with configurable proximity expansion zone (dashed bounding box visual).
- Real-time bounding box overlay at 100ms intervals drawing directly on the canvas, bypassing React state round-trips for minimal latency.
- The system correctly distinguishes a guide standing near wildlife (no alert) from a guide's hand reaching toward wildlife or flora (alert), directly addressing the false-positive problem documented in the project proposal.

5. Daily Progress Tracking

The table below records progress across the 6-week Sprint 1 period. Each session involved rotating pairs working 6-hour blocks, 3 days per week.

Week	Date (approx.)	Team Members	Tasks Completed	Hrs
Week 1	9–11 Mar	Terence, Emily, Miaw Fong	Project setup, Next.js scaffolding, Supabase project creation, database schema design, initial RLS policies	18
Week 2	16–18 Mar	Terence, Tammy, Emily, Miaw Fong	Google OAuth integration, OTP flow, JWT session management, middleware route protection, A2 RLS policy authoring	18
Week 3	23–25 Mar	Zi Yang, Young Fung	Training module CRUD API, Supabase Storage upload, HoD module editor UI, module versioning, module listing with progress	18
Week 4	30 Mar – 1 Apr	Brian, Zi Yang, Tammy, Emily, Miaw Fong, Young Fung	Quiz builder UI, quiz attempt UI, content player, role-based UI rendering, RLS unit testing, public placeholder pages	18
Week 5	7–9 Apr	Terence, Tammy, Zi Yang	AI dataset collection and Roboflow pipeline, YOLOv8n model training (72 epochs), ONNX export	18
Week 6	14–16 Apr	Terence, Tammy	ONNX Runtime Web integration, live inference loop, bounding box overlay, alert system, proximity detection tuning	18

6. Sprint Retrospective

6.1 What went well

- The AI monitoring prototype (D2–D4 + model training) was completed significantly ahead of schedule by Terence, Tammy, and Zi Yang, validating the technical architecture before Sprint 2 begins.
- Authentication (A1) and RBAC (A2) were completed cleanly in the first two weeks across the full team, providing a stable foundation for all role-specific feature development.
- Training module management (B1) is fully functional, completed entirely by Zi Yang, making it ready for HoD use from Sprint 2 onwards.
- Local model training on the RTX 5070 Ti eliminated Colab session limitations and significantly reduced the training iteration cycle.

6.2 What could be improved

- Task tracking was not consistently maintained in Trello during Sprint 1. The team will set up Trello with burndownfortrello.com at the start of Sprint 2 and update it daily.
- Notifications (C1–C3) were planned for Sprint 1 but not started. The dependency chain with B4–B7 must be accounted for in Sprint 2 scheduling.
- The flower class mAP50 (0.603) is the weakest. Additional flower training data should be collected and the model fine-tuned in Sprint 2.
- Service worker model caching (D1) was not completed, leaving the AI monitoring system not fully offline-capable. This is an explicit Sprint 2 acceptance criterion and must be prioritised.

6.3 Sprint 2 priorities

- Complete D1 service worker model caching to enable fully offline AI monitoring.
- Build D5–D7 evidence pipeline: clip capture, FFmpeg.wasm H.264 compression, Supabase Storage upload, offline queue, and background sync.
- Complete B4–B8 certification lifecycle: application, review, payment, badge issuance, senior nomination.
- Implement C1–C3 notification systems: email, SMS (Twilio), and in-app (Supabase Realtime).
- Set up Trello board and burndownfortrello.com at Sprint 2 kickoff and maintain daily.